DCS SHEET#1 FALL 2026 · STUDY GUIDE

1. Define Distributed Computer System DCS? What are the pros and cons of Distributed system over centralized system?

A:

DCS: It is one in which components located at networked computers communicate and coordinate their actions only by passing messages.

Pros over a centralized system:

Distributed systems aim to improve performance, reliability, scalability, and resource sharing by leveraging the collective power of interconnected devices.

1 · Foundation

1. **Concurrency:** Multiple nodes operate concurrently, performing tasks simultaneously. This parallelism enhances the system's overall performance and throughput.
2. **Scalability:** Easy to scale by adding more nodes to the system. System capacity and performance improve with the addition of resources.

2 · Access and resources

3. **Resource Sharing:** Nodes share resources such as processing power, storage, and data. This enables more efficient utilization of resources.

3 · Management and protection

4. **Fault Tolerance:** Designed to handle failures gracefully. Redundancy and replication ensure the system remains operational even if some nodes fail.

4 · Multi-user safety net

5. **Transparency:** The complexity of the distributed system is hidden from users. Users interact with the system as if it were a single entity.

Cons over a centralized system:

1 · Foundation

- **Latency:** Potentially higher latency due to network communication.

3 · Management and protection

- **Management:** Complex, requires coordination and management of many nodes.
- **Failure handling:** It is difficult in DCS because the failure is partial.

Advantages of distribution vs. advantages of centralization:

Advantages of DISTRIBUTION	Advantages of Centralization
1) Technological changes	1) Economy of scale
2) User needs	2) More powerful capabilities
3) Modularity and simpler software	3) Operating costs
4) Flexibility and Extension	4) Staff Satisfaction

5) Availability and integrity	5) Local programming and development
6) Performance	6) Communication subsystem
7) Local control	7) Stretching the state of the art

Study notes

Acronym: Pros **C-S-R-F-T**, cons **L-M-F**, table: 7 against 7, numbered.

Trail: An office building: the crew works at the same time (Concurrency), you can add floors (Scalability), everyone shares the printer (Resource sharing), a generator keeps the lights on (Fault tolerance), and visitors see one company (Transparency). The price: messages take time (Latency), many floors to run (Management), and half the building can fail (Failure handling).

If you forget: Walk the four places: how fast and big, what is shared, what protects it, what the users see.

Link: Q7 has the same Latency and Management rows; Q20 lists the cluster version of these pros.

Test yourself: Why is failure handling harder in a DCS than in a centralized system?

2. Define data centers and mention its applications?

A:

Data Centers DCs: They are the physical facilities that house the IT equipment and infrastructure necessary to store, process, and distribute vast and ever-growing volume of data.

Its applications:

1. Servers
2. Storage systems
3. Network devices
4. Cybersecurity devices
5. Other infrastructure components

Study notes

Acronym: **S-S-N-C-O**: Servers, Storage, Network, Cybersecurity, Other.

Trail: Walk into the hall: servers think, storage remembers, the network talks, security guards the door, the rest supports.

If you forget: Store, process, distribute: what does each job need?

Test yourself: Write the definition with its three verbs.

3. Mention Data Centers components?

A:

Main components of a datacenter:

1 · The building

1. Physical Suitable Site
2. Racks and enclosures
3. Power Distribution Units PDUs
4. Cooling systems

2 · The IT inside

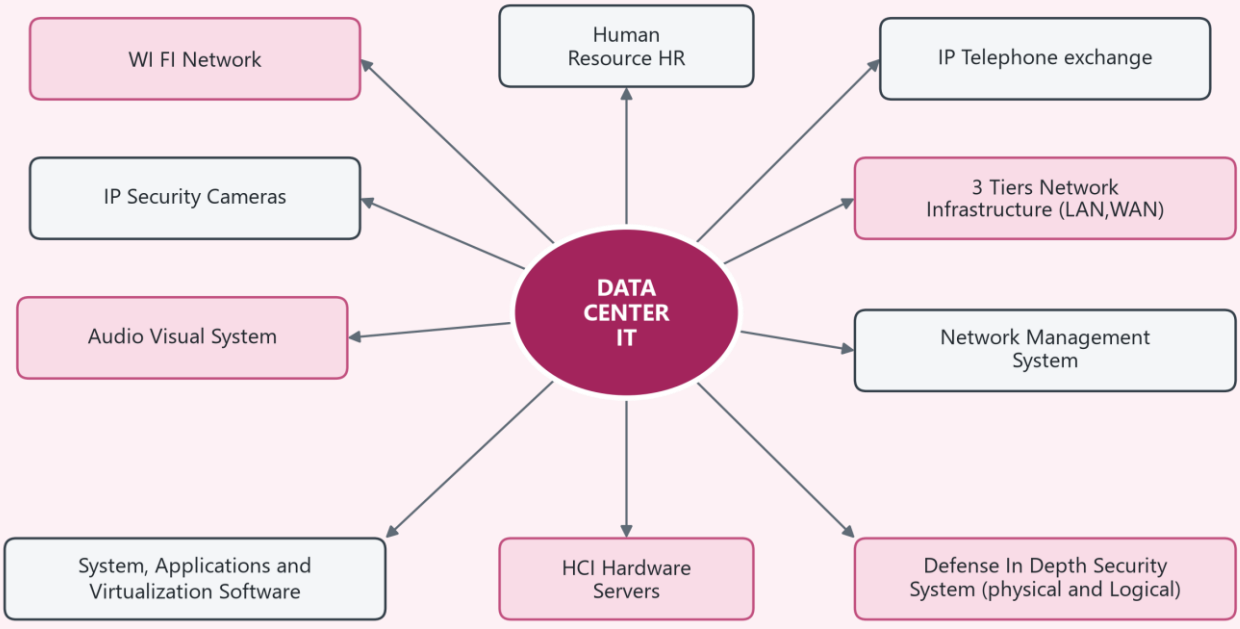
5. Computer resources
6. Storage systems
7. Network equipment
8. Software and virtualization

3 · Protection and control

9. Fire suppression systems
10. Security systems
11. Data Center Management

4 · People

12. Human Resources



Study notes

Acronym: Building **S-R-P-C**, IT **C-S-N-S**, protection **F-S-M**, then **H**umans.

Trail: Building a data center: buy the site, put up racks, bring power, cool it; fill it with computers, storage, network and software; protect it from fire and intruders and manage it; then hire the people.

If you forget: Build it, fill it, protect it, staff it.

Link: Q2 is the short list of the same hall.

Test yourself: Name the four things that come before any computer is installed.

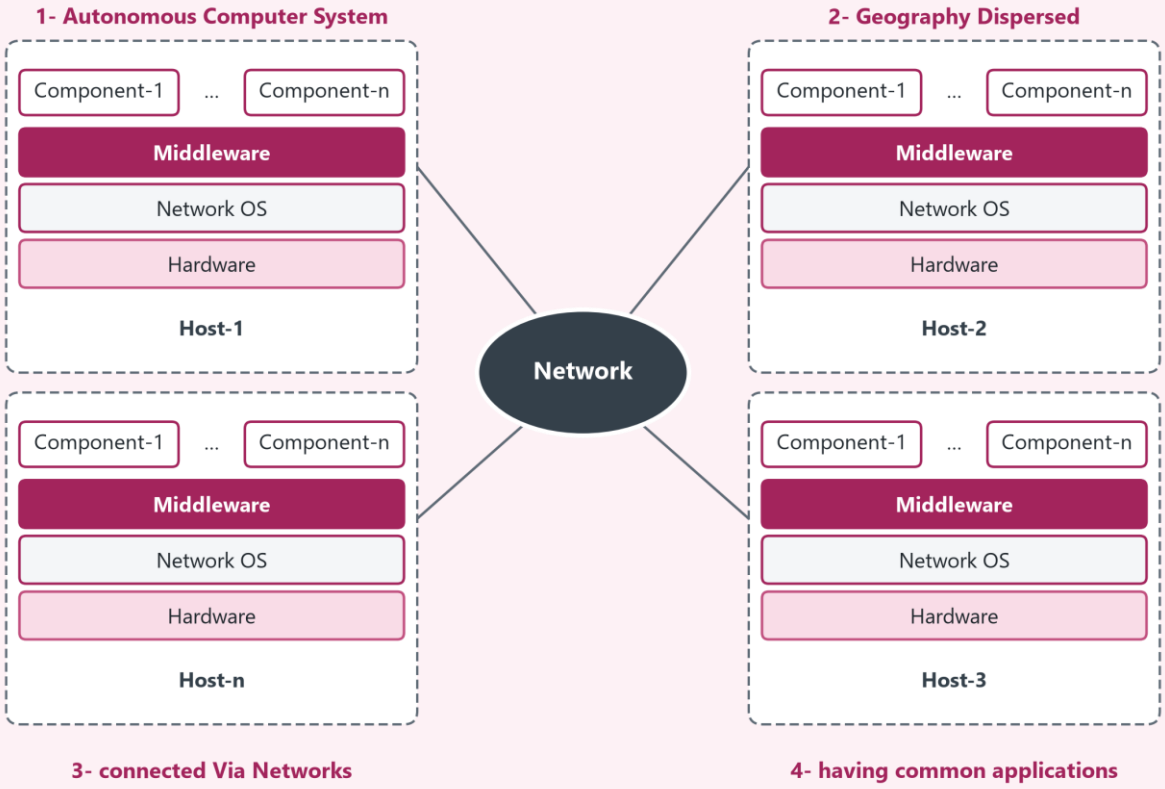
4. Explain the key components of distributed computing systems using drawing?

A:

DCS Components: Group of autonomous computing systems + geographically dispersed + connected via LAN/WAN + use messages to communicate voice/video/data and to run distributed applications

A Distributed Computer System is a system with:

1. Group of autonomous systems
2. Geographically dispersed
3. Connected via a network
4. Having common/shared applications



Components of Distributed Software:

- Distributed operating systems
- Middleware
- Distributed applications

Study notes

Acronym: **A-G-C-S**: Autonomous, Geographic, Connected, Shared applications.

Trail: Four friends in four cities (autonomous, dispersed) on one group chat (network) playing one game (shared app).

If you forget: The definition line already holds the four parts in order; the software list goes up the layers.

Link: Q6 draws the same software layers; Q17 explains the middleware layer.

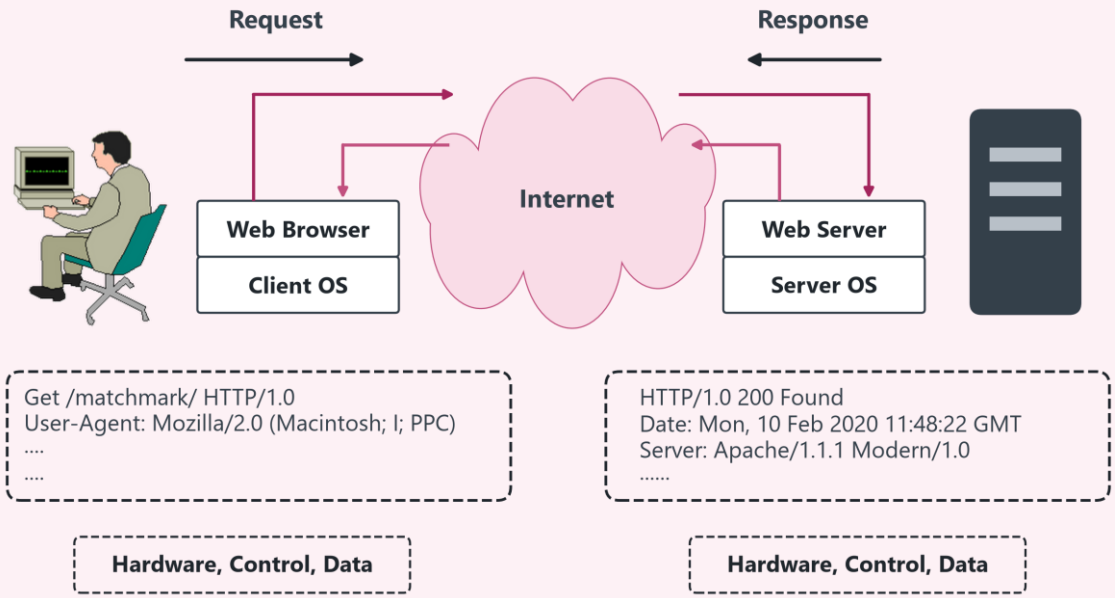
Test yourself: Which three kinds of software make up distributed software?

5. With the help of a diagram explain the working of distributed systems.

A:

DCS: It is a collection of autonomous computer systems that are physically separated but are connected by a computer network that is equipped with distributed system software.

- **Requests:** Sent in messages from clients to a server.
- **Replies:** Sent in messages from the server to the clients.



Study notes

Acronym: Req / Rep: requests go client to server, replies come back.

Trail: A restaurant: the client orders, the server cooks and replies, the street between them is the network.

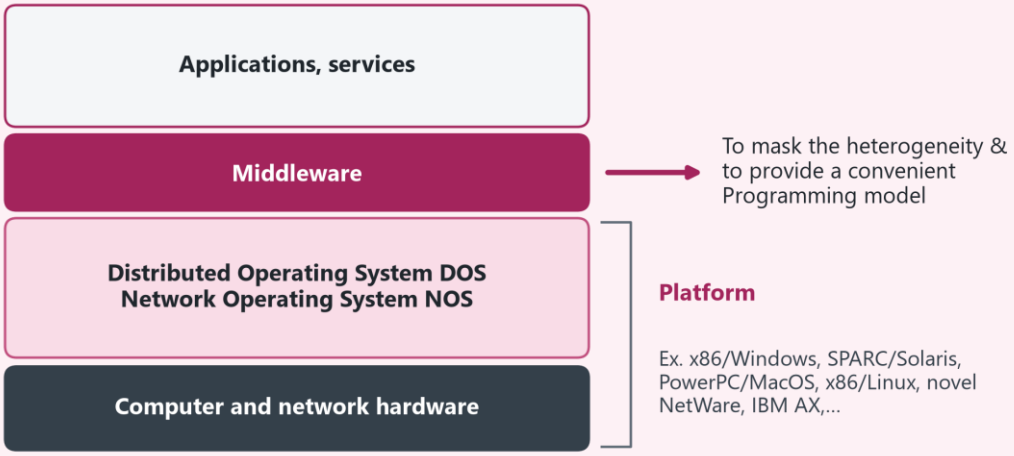
If you forget: Draw the browser, the server and two arrows.

Link: Q4 has the same definition words (autonomous, connected).

Test yourself: Who sends requests and who sends replies?

6. Draw the DCS software architecture and compare between distributed operating system and network operating system?

A:



Aspect	NOS	DOS
Definition	A NOS is an operating system that is expected to support networking functionalities.	DOS is responsible for coordinating the computers/nodes that are connected and operates in cooperation like a single large machine.
Examples	MS SERVER, IBM AIX, ORACLE SOLARIS, LINUX SERVER, UNIX	MS AZURE, GOOGLE FILE SYSTEM GFS, LOCUS, ORACLE SOLARIS
Advantages	Scalability Resource Sharing Centralized Control NOS Security	Scalability Resource Utilization Fault Tolerance DOS Security Transparency

Study notes

Acronym: NOS **S-R-C-S**, DOS **S-R-F-S-T**: both start with Scalability and Resources, DOS ends with Transparency.

Trail: NOS is roommates sharing a fridge with one landlord (control); DOS is one big office where nobody knows which desk did the work (transparency).

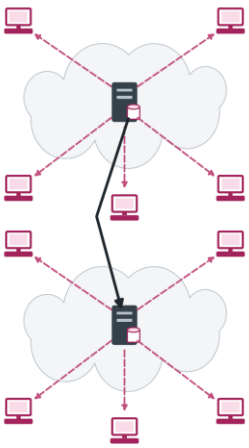
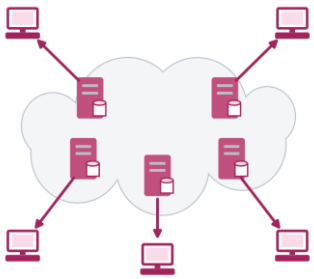
If you forget: Same chain as Q8: scale, resources, protection, users.

Link: Q8 and Q11 use the same chain.

Test yourself: Which advantage does DOS have that NOS does not, and the reverse?

7. Compare between centralized, decentralized, and distributed computing systems?

A:

Aspect	Centralized Systems	Decentralized Systems	Distributed Systems
Definition	Single central server controls and manages all operations.	Multiple nodes with independent control, no central authority.	Multiple interconnected nodes working together as a single system.
1 · Foundation			
Performance	Can be high initially but may degrade with increased load.	Generally good, performance improves with more nodes.	High performance due to parallel processing and resource sharing.
Latency	Lower latency, as operations are managed centrally.	Can vary, depends on the distance between nodes.	Potentially higher latency due to network communication.
Scalability	Limited scalability, can become a bottleneck.	More scalable, can add nodes independently.	Highly scalable, can add more nodes to distribute the load.
2 · Access and resources			
Resource Utilization	Central server resources are heavily utilized.	Resources are spread across multiple nodes.	Efficient resource sharing across nodes.
3 · Management and protection			
Control	Centralized control with a single point of management.	Distributed control, each node operates independently.	Shared control, nodes collaborate to achieve common goals.
Management	Easier to manage centrally.	More complex, requires managing multiple nodes.	Complex, requires coordination and management of many nodes.
Single Point of Failure	High risk; if the central server fails, the whole system fails.	Reduced risk; failure of one node does not impact the entire system.	Reduced risk; designed for fault tolerance and redundancy.
Drawing			

Study notes

Acronym: **P-L-S | R | C-M-F** between Definition and Drawing.

Trail: One boss, many bosses, one team: how fast they work and wait, how they share the tools, who controls and manages, and what happens when one falls.

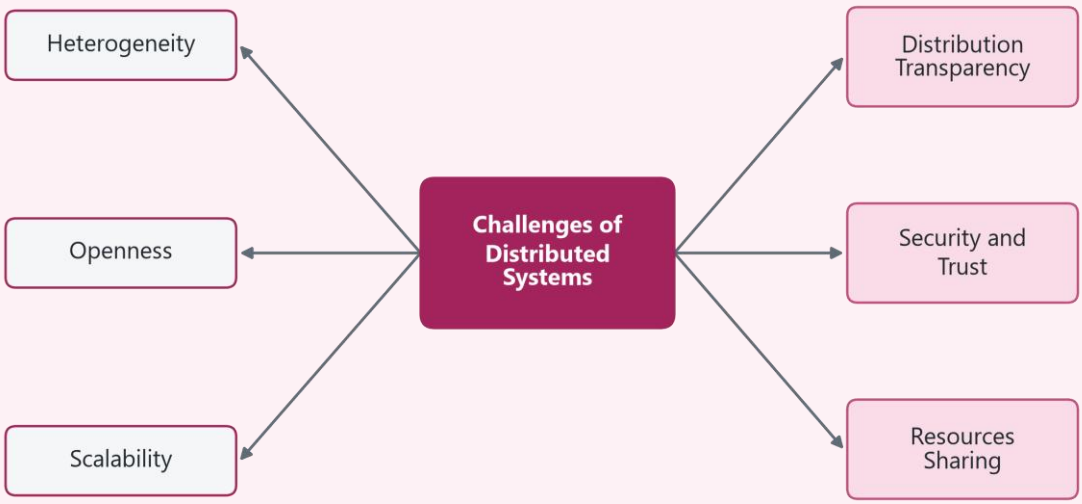
If you forget: Performance first, failure last; centralized is always the easy but risky column.

Link: Q1 cons repeat the Latency and Management rows of the distributed column.

Test yourself: Which system has the highest single-point-of-failure risk, and why?

8. Identify the various challenges in the distributed system.

A:



1 · Foundation

- **Scalability:** A system is described as scalable if it will remain effective when there is a significant increase in the number of resources and the number of users.

2 · Access and resources

- **Heterogeneity:** DCSs enable users to access services and run applications over a heterogeneous collection of computers and networks.
- **Openness:** It is the characteristic that determines whether the system can be extended and re-implemented in various ways.
- **Resources Sharing:** Decentralized scheduler organization negates the limitations of centralized organization with respect to fault-tolerance, scalability, and autonomy.

3 · Management and protection

- **Security and Trust:** The decentralized organization of distributed systems raises serious challenges in domains of security and trust management.

4 · Multi-user safety net

- **Distribution Transparency:** It is the concealment from the user and the application programmer of the separation of components in a distributed system.

Study notes

Acronym: **S-H-O-R-S-T**: Scalability, Heterogeneity, Openness, Resources sharing, Security, Transparency.

Trail: A growing shipping company: it must scale, carry any kind of package (heterogeneity), accept new partners (openness), share trucks (resources), guard the cargo (security), and the customer sees one parcel service (transparency).

If you forget: SHORST: one letter per challenge, in the chain order.

Link: Q13 (scalability), Q14 (resource sharing), Q15 (openness), Q16 (fault tolerance) and Q9 (transparency) each take one.

Source note: The slide's "Resources Sharing" line is about scheduler organization; the full definition is in Q14.

Test yourself: Which challenge decides whether the system can be extended and re-implemented?

9. Explain what is meant by (distribution) transparency, and give examples of different types of transparency.

A:

Distribution Transparency: It ensures that the DCS should be perceived as a single entity by the users or the application programmers rather than a collection of autonomous systems.

Types of transparency:

Transparency	Description
1 · Where it is	
Location	Hide where a resource is located, users cannot tell where hardware and software resources such as Servers, printers, files, data bases are located.
Migration	Hide that a resource may move to another location
Relocation	Hide that a resource may be moved to another location while in use
2 · How it is reached and copied	
Access	Hide differences in data representation and how a resource is accessed
Replication	Hide that a resource is replicated, OS can make additional copies of files and resources without users noticing.
3 · Protection	
Failure	Hide the failure and recovery of a resource
4 · Many users	
Concurrency	Hide that a resource may be shared by several competitive users, The users are not aware of the existence of other users.
Concurrency Transparency	Automatic use of parallelism without having to program explicitly.

Study notes

Acronym: **L-M-R | A-R | F | C-C**

Trail: A library book: where it is (location), it moves (migration), it moves while you read it (relocation); how you get it and its copies (access, replication); it gets lost and replaced (failure); others read it too (concurrency).

If you forget: Place, use, protection, other users.

Link: Q8: Distribution Transparency is the last challenge there.

Source note: The last row is named Concurrency Transparency on the slide, but its description is about parallelism.

Test yourself: What is the difference between migration and relocation transparency?

10. Explain the differences between homogeneous and heterogeneous systems?

A:

Heterogeneity: It is applied to the network, computer hardware, operating system, and implementation/development of different developers.

Subsystems	Homogeneous Systems	Heterogeneous Systems
1 · Foundation		
Increased Performance	Simple	Difficult
Growth	Simple	Difficult
2 · Access and resources		
Computer hardware	Same	Variant/Different
Operating systems	Same	Variant/Different
Programming languages	Same	Variant/Different
Networks	Same	Variant/Different
3 · Management and protection		
Design and Management	Simple	Difficult
4 · Multi-user safety net		
Implementations by different developers/vendors	Same	Variant/Different

Study notes

Acronym: **P-G | H-O-L-N | D | I**

Trail: Same team vs mixed team: performance and growth are simple only when all match; hardware, OS, languages and networks are the parts that differ; design is harder; and different vendors mean many builders.

If you forget: Homogeneous column = Same/Simple, heterogeneous = Different/Difficult.

Link: Q11: middleware exists to mask this heterogeneity.

Test yourself: Name the four subsystems listed in the heterogeneity definition.

11. List the functions of middleware.

A:

1 · Foundation

- Extends over multiple machines, and offers each application the same interface
- Bridges gap between low-level OS communications and programming language abstractions

2 · Access and resources

- Provides a uniform computational model for use by the programmers of servers and distributed applications to communicate between homogeneous and heterogeneous systems.
- Provides common programming abstraction and infrastructure for distributed applications

3 · Management and protection

- Hides complexity and heterogeneity of distributed system

4 · Multi-user safety net

- Enables applications and end-user to interact with each other across a heterogeneous distributed system by masking the heterogeneity of the networks, hardware, OS, and programming languages

> **Study notes**
>
> **Acronym: E-B-U-C-H-E**
>
> **Trail:** A translator between countries: works across borders (extends), bridges two languages, gives one grammar (uniform model) and one phrasebook (common abstraction), hides the accents, and lets everyone talk (enables).
>
> **If you forget:** Middleware sits between the OS and the application: what does each side need?
>
> **Link:** Q17 is the same list without the last line; Q10 is the heterogeneity it masks.
>
> **Test yourself:** What does middleware mask?

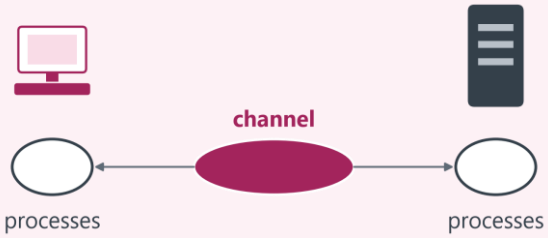
12. Give examples for security threats and security protection.

A:

Encryption: It protects shared resources and keeps sensitive information secret when transmitted.

Security Threats include:

1. Threats to the processes
2. Threats to the servers
3. Threats to the communication channel
4. Denial of service DOS attacks



Examples of Security Tools:

1. **Access rights** – specify who is allowed to perform the operations of an object
2. **Principal** – associated with each session/association and each result the authority on which it is issued. A principal may be a user or a process
3. **Encryption and authentication**- are used to build secure channels as a service layer on top of existing communication services.

> **Study notes**
> **Acronym:** Threats **P-S-C-D**, tools **A-P-E**.
> **Trail:** Two processes talk over a channel: an attacker can hit either process, the server, the channel, or flood it (DoS). Guards: who may do what (access rights), who is asking (principal), and a locked pipe (encryption).
> **If you forget:** Follow the drawing from left to right for the threats.
> **Test yourself:** Which tool builds secure channels?

13. What is meant by scalability challenge and how to achieve it?

A:

Scalability: The system should remain efficient even with a significant increase in the number of users and resources connected.

How to achieve it:

- Adding more and/or faster processors.
- Components should not need to be changed when scale of a system increases.

> **Study notes**
> **Acronym:** **More + No change**
> **Trail:** A restaurant that adds tables without rebuilding the kitchen.
> **If you forget:** Two answers: add processors, don't change components.
> **Link:** Q8 first challenge.
> **Test yourself:** Define scalability in one line.

14. What is meant by resources sharing challenge and how to achieve it?

A:

Resources Sharing: Resources in a distributed system are physically encapsulated within computers and can only be accessed from other computers by means of communication.

How to achieve it:

- Each resource must be managed by a program that offers a communication interface enabling the resource to be accessed and updated reliably and consistently.
- The term service is used for a distinct part of a computer system that manages a collection of related resources and presents their functionality to users and applications.
- Decentralized scheduler organization negates the limitations of centralized organization with respect to fault-tolerance, scalability, and autonomy.

> **Study notes**
>
> **Acronym:** **Program, Service, Scheduler**
>
> **Trail:** A shared kitchen: each appliance has a manager with a door (interface), the manager's job is a service, and no single chef schedules everything.
>
> **If you forget:** Resources are locked inside computers, so communication is the only way in.
>
> **Link:** Q8 uses the scheduler sentence for Resources Sharing.
>
> **Test yourself:** Why can resources only be reached by communication?

15. What is meant by openness challenge and how to achieve it?

A:

Openness: It is determined by the degree to which new resource-sharing services can be made available to the users.

- Open systems are characterized by the fact that their key interfaces are published.

How to achieve it:

- It is based on a uniform communication mechanism and published interface for access to shared resources.
- Detailed interfaces of components need to be published.
- New components have to be integrated with existing components.
- Differences in data representation of interface types on different processors (of different vendors) have to be resolved.
- Openness cannot be achieved unless the specification and documentation of the key software interfaces of the components of a system are made available to software developers.

> **Study notes**
> **Acronym:** **Publish, Integrate, Resolve**
> **Trail:** A plug standard: publish the socket shape, new devices must fit the old sockets, and adapters fix voltage differences.
> **If you forget:** Openness lives on published interfaces.
> **Link:** Q8 Openness; Q10 data representation differences.
> **Test yourself:** What can openness not be achieved without?

16. What is meant by fault tolerance challenge and how to achieve it?

A:

Fault Tolerance: Distributed systems must maintain availability even at low levels of hardware/software/network reliability.

- **Failure handling:** It is difficult in DCS because the failure is partial.

Fault tolerance is achieved by:

- Recovery
- Redundancy

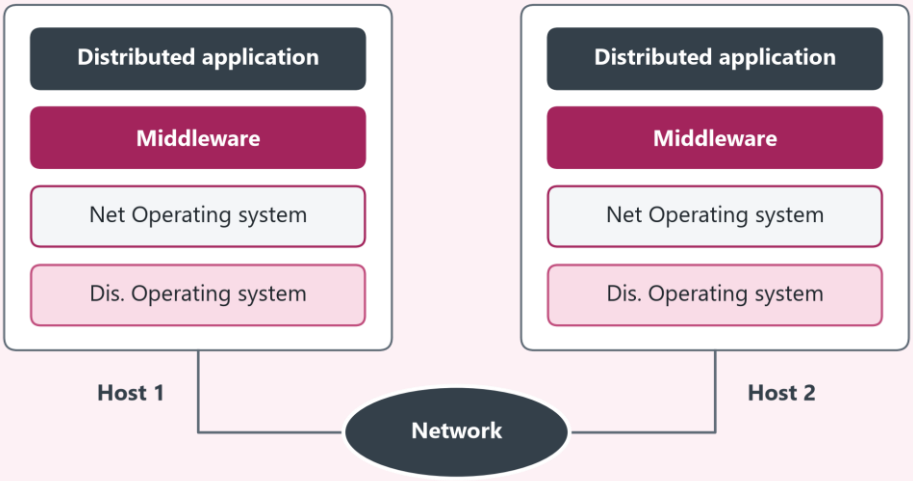
> **Study notes**
> **Acronym:** **R-R**: Recovery, Redundancy.
> **Trail:** A spare tire (redundancy) and a mechanic (recovery).
> **If you forget:** Partial failure is the problem; two R's are the fix.
> **Link:** Q1 cons, Failure handling; Q20 redundancy.
> **Test yourself:** Why is failure handling difficult in a DCS?

17. Explain and List the functions of middleware.

A:

Middleware: It is a set of services that enables applications and end-user to interact with each other across a heterogeneous distributed system by masking the heterogeneity of the networks, hardware, OS, and programming languages

- It is Layer between (DOS and NOS) and distributed applications



Functions of middleware:

1 · Foundation

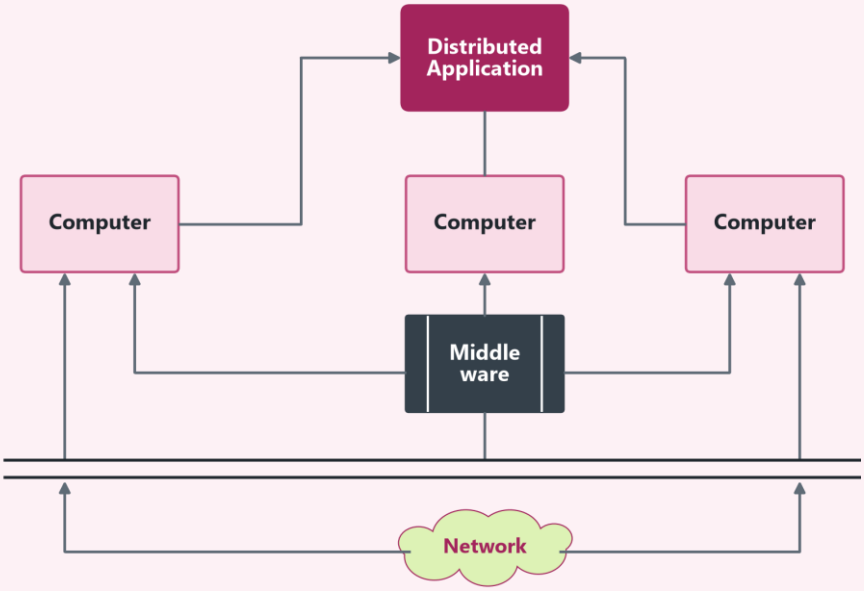
- Extends over multiple machines, and offers each application the same interface
- Bridges gap between low-level OS communications and programming language abstractions

2 · Access and resources

- Provides a uniform computational model for use by the programmers of servers and distributed applications to communicate between homogeneous and heterogeneous systems.
- Provides common programming abstraction and infrastructure for distributed applications

3 · Management and protection

- Hides complexity and heterogeneity of distributed system



Study notes

Acronym: **E-B-U-C-H**

Trail: The Q11 translator, without the last line.

If you forget: Definition, layer, then the list from Q11.

Link: Q11 is the same list plus Enables.

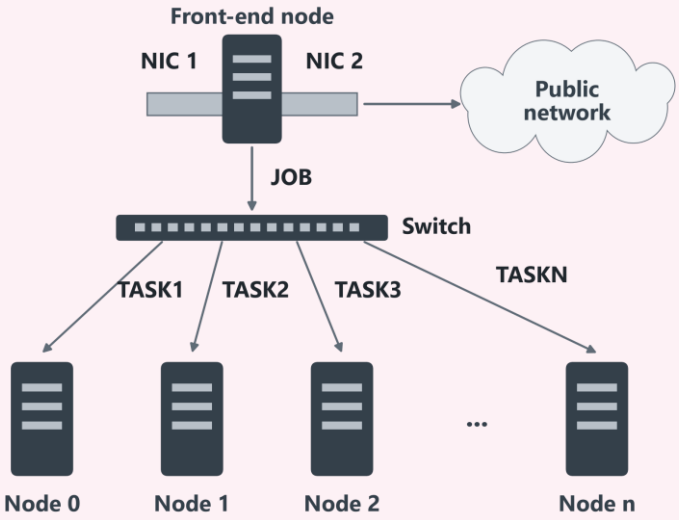
Test yourself: Between which two layers does middleware sit?

18. Explain and List the functions of clustering computing.

A:

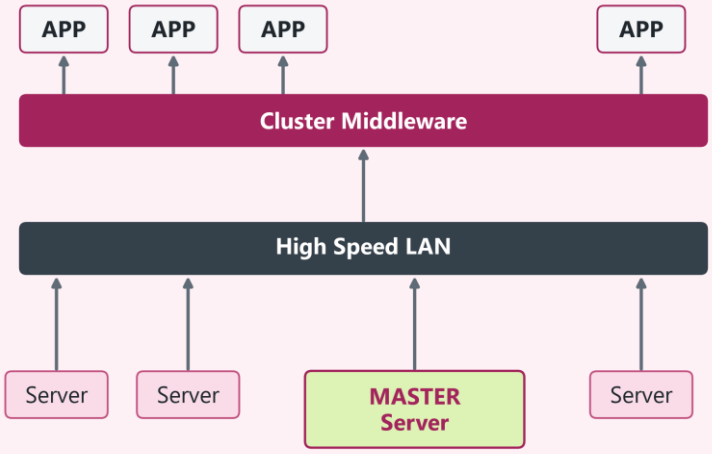
Computing Cluster: It is a set of connected computers that work together so closely that in many respects they function as a single computer.

- **Idea:** To take a large computational task, break it down into discrete parts, and distribute those parts across different computers to be processed simultaneously.
- The components of a cluster are connected to each other through fast LANs.



Cluster consists of:

- Nodes (master + computing)
- Network
- OSs
- Cluster middleware which permits compute clustering programs to be portable to a wide variety of clusters



Functions of cluster computing:

- Improve performance and/or availability over that provided by a single computer
- Enables scalable parallel and distributed computing in both science and business applications.

Goal: To provide a range of cloud services, including high-performance computing capabilities, mass storage (e.g. data centers), and richer application services such as web search.

Study notes

Acronym: Cluster = **N-N-O-M** (Nodes, Network, OSs, Middleware).

Trail: A big puzzle: break it into pieces, hand them out on a fast LAN, everyone works at once.

If you forget: Break, distribute, process simultaneously.

Link: Q20 benefits and Q21 types of clusters.

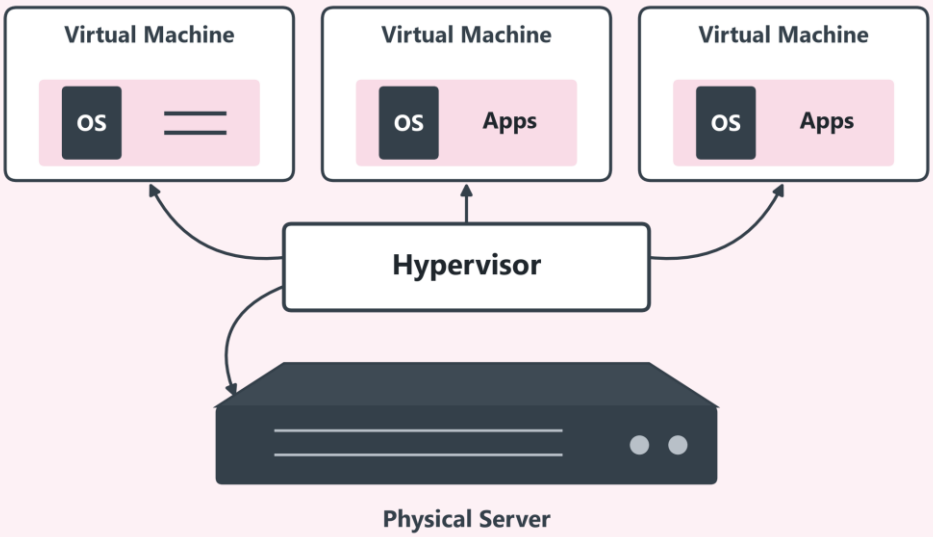
Test yourself: What connects the components of a cluster?

19. Explain and List the functions of virtualization.

A:

Virtualization: It is the ability to run multiple OSs on a physical system and share the hardware, software, storage, Desktop, printing, and network resources

- It is a technique of partitioning the resources of a single computing platform into multiple segregated, virtualized, execution environments.
- Each environment runs independently of the other, thus allowing multiple OSs to run on the same hardware.
- **Guest:** Each execution environment.
- **Host:** The computing platform on which they execute.
- **Hypervisor or Virtual Machine Monitor (VMM):** The software enabling these multiple execution environments.



Functions of virtualization:

- Enables sharing of resources much easily, it helps in increasing the degree of hardware level parallelism
- Helps in sandboxing capabilities or migrating different kinds of applications that in turn enables flexible allocations and utilization of the resources.

Study notes

Acronym: G-H-H: Guest, Host, Hypervisor.

Trail: An apartment building: one building (host), many flats (guests), one building manager (hypervisor).

If you forget: Partition one machine into many environments.

Test yourself: What is the software that enables multiple execution environments called?

20. What are the benefits of a computer cluster?

A:

1 · Foundation

- Much more cost effective than single computers of comparable speed or availability
- Significant increase in processing speed and efficiency, allowing tasks to be completed more quickly and effectively than with a single computer.
- Improve performance and/or availability over that provided by a single computer
- Scalable parallel and distributed computing in both science and business applications.

3 · Management and protection

- Provides redundancy
- Eliminate single point of failure

> **Study notes**
> **Acronym:** **$-S-P-S | R-E**
> **Trail:** Buying a team instead of one star: cheaper, faster, better, grows with you, and a spare player so no one is irreplaceable.
> **If you forget:** Cost first, failure last.
> **Link:** Q1 pros; Q21 HA clusters repeat redundancy and single point of failure.
> **Test yourself:** Why is a cluster more cost effective than a single computer?

21. What are the types of cluster computing?

A:

High availability Clusters (HA) (Linux):

- Mission critical applications
- High availability /Failover clusters are implemented for the purpose of improving the availability of services which the cluster provides
- Provides redundancy
- Eliminate single point of failure

Network Load balancing Clusters:

- Operate by distributing a workload over multiple backend nodes
- The cluster will be configured with multiple redundant load-balancing front ends
- All available servers (Web servers, mail servers, ftp servers,….) process requests

Science Clusters:

- Science cluster is a computer design that uses parallel and cluster processing across multiple computers to create a cheap powerful supercomputer
- It has two types of computers, a master/front/service computer and node computers
- When a large problem or set of data is given, the master computer first runs a program that breaks the problem into discrete pieces, it then sends a piece to each node to compute
- As nodes finish their tasks, the master computer sends more pieces to them until the entire problem has been computed

> **Study notes**
> **Acronym: HA - NLB - Science**
> **Trail:** A hospital (high availability), a supermarket with many tills (load balancing), a lab sharing one huge puzzle (science).
> **If you forget:** Stay up, share load, compute.
> **Link:** Q20 redundancy and single point of failure; Q18 the master and node idea.
> **Test yourself:** Which cluster type uses a master computer that breaks the problem into pieces?

GOOD LUCK

Quick review

Q	Question	Remember
1	Define Distributed Computer System DCS? What are the pros and cons of Distributed system over centralized system?	DCS = messages only. Pros C-S-R-F-T, cons L-M-F.
2	Define data centers and mention its applications?	House IT to store, process, distribute. S-S-N-C-O.
3	Mention Data Centers components?	12 parts: build (4), fill (4), protect (3), staff (1).
4	Explain the key components of distributed computing systems using drawing?	A-G-C-S; software: DOS, middleware, applications.
5	With the help of a diagram explain the working of distributed systems.	Clients send requests, server sends replies.
6	Draw the DCS software architecture and compare between distributed operating system and network operating system?	NOS = networking OS, DOS = one large machine.
7	Compare between centralized, decentralized, and distributed computing systems?	Rows P-L-S-R-C-M-F; centralized easy but risky.
8	Identify the various challenges in the distributed system.	SHORST.
9	Explain what is meant by (distribution) transparency, and give examples of different types of transparency.	Single entity. L-M-R, A-R, F, C-C.
10	Explain the differences between homogeneous and heterogeneous systems?	Homo = same/simple, hetero = different/difficult.
11	List the functions of middleware.	E-B-U-C-H-E; masks heterogeneity.
12	Give examples for security threats and security protection.	Threats P-S-C-D, tools A-P-E.
13	What is meant by scalability challenge and how to achieve it?	Efficient with more users/resources; add processors, no change.
14	What is meant by resources sharing challenge and how to achieve it?	Encapsulated resources, interface program, service.
15	What is meant by openness challenge and how to achieve it?	Published interfaces, uniform communication.
16	What is meant by fault tolerance challenge and how to achieve it?	Keep available; recovery + redundancy.
17	Explain and List the functions of middleware.	Layer between DOS/NOS and apps; E-B-U-C-H.
18	Explain and List the functions of clustering computing.	Work as a single computer; N-N-O-M.
19	Explain and List the functions of virtualization.	Many OSs on one system; guest, host, hypervisor.
20	What are the benefits of a computer cluster?	Cost, speed, performance, scalable, redundancy, no SPOF.
21	What are the types of cluster computing?	HA, network load balancing, science.

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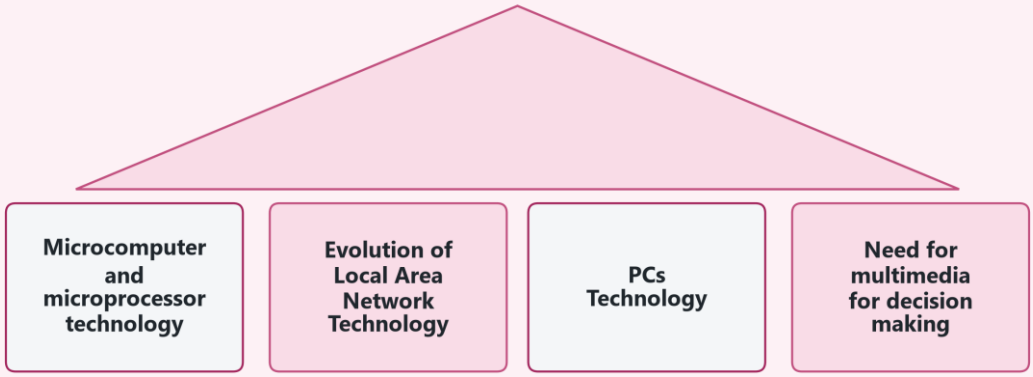
1. Explain the technologies that move the computer systems from centralized to distributed.

A:

Four advances in technology began to change from centralized to distributed computer systems:

1. **Microcomputer and microprocessor technology:** The development of powerful microprocessors.
2. **Evolution of Local Area Network Technology:** The invention of high-speed LAN and WAN computer networks.
3. **PCs Technology:** Miniaturization of computer systems with the smartphone as the most impressive outcome.
4. **Need for multimedia for decision making:** Decision makers are asking for multimedia (text, audio, video, graphs) for decision making.

Result: A computing system composed of many networked computers, be they large or small.



Study notes

Acronym: M-L-P-M: Microprocessors, LAN, PCs, Multimedia.

Trail: Chips got strong, cables got fast, computers got small, and managers wanted videos: so one big computer became many.

If you forget: Chip, cable, device, content.

Link: Q2, Q3 and Q4 each explain one of the first three.

Test yourself: What was the result of the four advances?

2. Mention the roles of microprocessor/minicomputer technology in the evolution of DCS?

A:

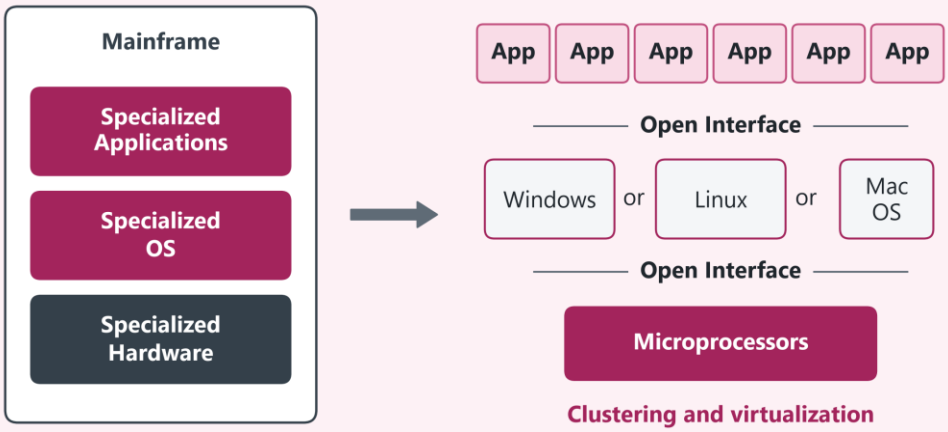
Roles:

1 · Foundation

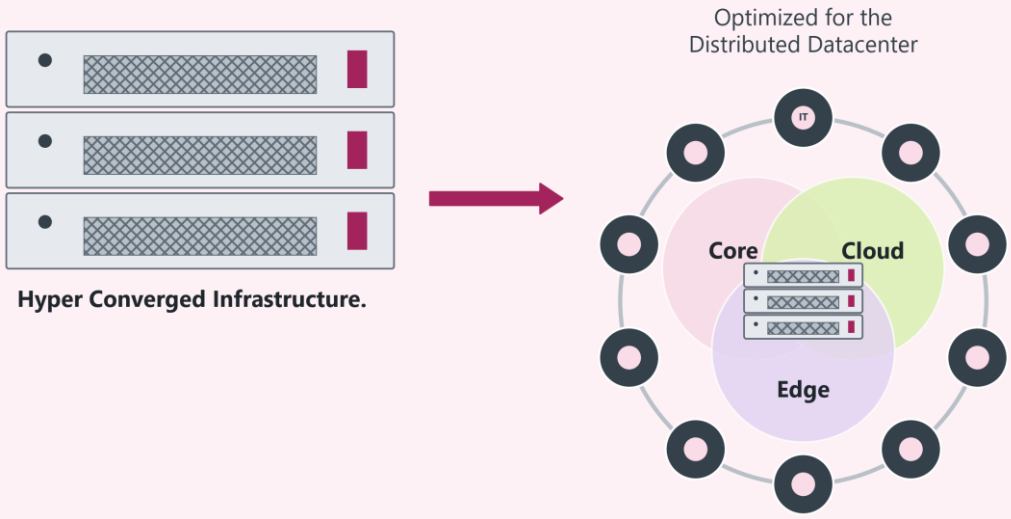
- Current generation of machines have the computing power of the mainframes deployed 30 or 40 years ago, but for 1/1000th of the price or less.
- 16-, 32-, and 64-bit CPUs became common.
- With multicore CPUs, we now are facing the challenge of adapting and developing programs to exploit clustering and parallelism.

2 · Access and resources

- Transformation of the vertical mainframe market to a horizontal marketplace with open interfaces and multiple options available at every level.



- **Hyper-converged infrastructure HCI:** It is a hardware/software-defined IT infrastructure that virtualizes all of the elements of conventional "hardware-defined" systems.



Study notes

Acronym: **$-B-M-T**, then HCI.

Trail: A mainframe that cost a fortune now fits a cheap chip, chips went 16 to 64 bits and multicore, and the market turned from one vendor's tower to open shelves.

If you forget: Price, bits, cores, market.

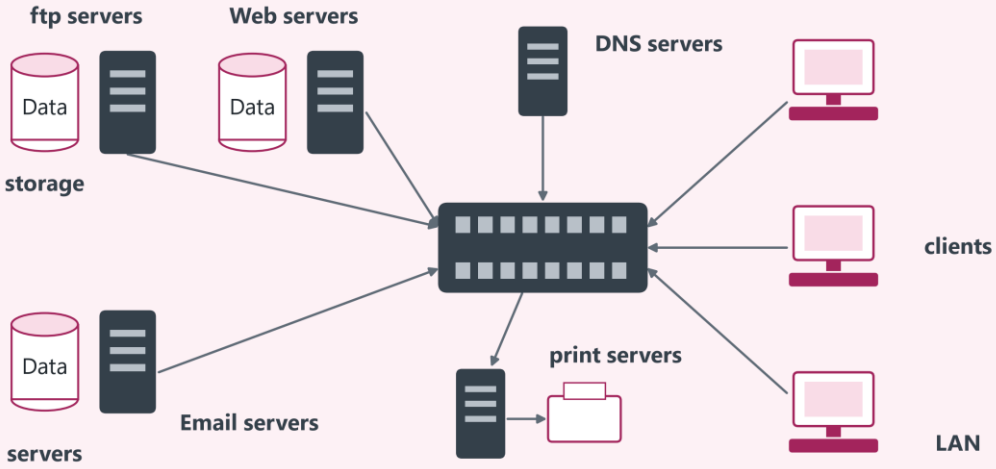
Link: Q1 first advance.

Test yourself: What does HCI virtualize?

3. Mention the roles of local area and wide area networks technology in the evolution of DCS?

A:

- **LANs:** They allow thousands of machines within a building or campus to be connected in such a way that small amounts of information can be transferred in a few microseconds or so, and larger amounts of data can be moved between machines at rates of 10^{12} bps.
- **WANs:** They allow hundreds of millions of machines all over the earth to be connected at speeds varying from 10^4 to 10^9 bps, and sometimes even faster.



Study notes

Acronym: **LAN** thousands, microseconds; **WAN** hundreds of millions, 10^4 to 10^9 bps.

Trail: A campus is a house party (thousands, fast), the earth is a world tour (millions, slower).

If you forget: Compare how many machines, how far, how fast.

Link: Q1 second advance.

Source note: 10^12 bps for a LAN is far above common LAN speeds (10^9 to 10^10 bps); write the slide value.

Test yourself: Over which speed range do WANs connect machines?

4. Mention the roles of the PC technology in the evolution of DCS?

A:

- **Smartphones:** Packed with powerful PCs, sensors, lots of memory, and a powerful CPU, these devices are nothing less than full-fledged computers.
- Plug computers and other so-called Nano computers are finding their way to the market.
- **Plug computers:** Often the size of a power adapter, can often be plugged directly into an outlet and offer near-desktop performance.
- **PC desktop computers and notebook:** They can perform all of its input, processing, output, and storage activities by itself.
- **Desktop computer:** It is designed so the system unit, input devices, output devices, and any other devices fit entirely on or under a desk or table.
- **Notebook computer:** It is a portable PC small enough to fit on a lap.

Study notes

Acronym: S-N-P | D-D-N: Smartphones, Nano, Plug; PC, Desktop, Notebook.

Trail: From the pocket to the plug to the desk to the lap.

If you forget: Small devices first, then the PC family.

Link: Q5 continues with the PC configurations.

Test yourself: Which computers are the size of a power adapter?

5. Mention the different applications/configurations of the PC in the evolution of DCS?

A:

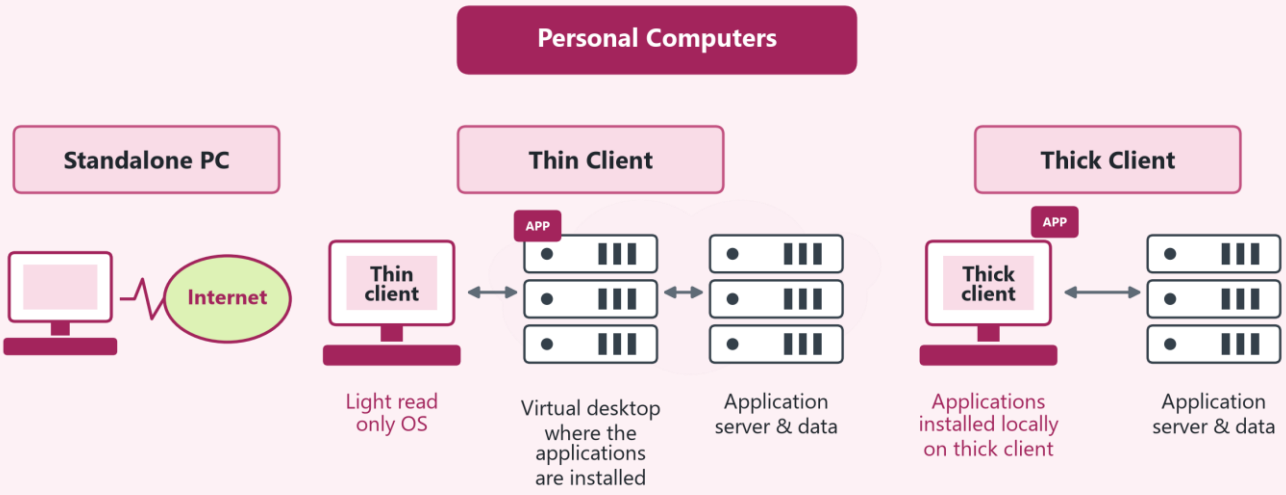
Notebook and desktop computers are used at home or in the office to perform application software-related tasks or to access the Internet.

Variations of desktop computers include:

- **Tower models:** Tall computers that can sit vertically on the floor.
- **All-in-one computers:** Computers that combine monitor and system unit into a single device.
- **Workstations:** More expensive and powerful computers designed for work that requires intense calculation and graphics capabilities.

Personal Computers:

1. Dumb Terminals
2. Emulation Mode
3. Diskless PC
4. Standalone PC
5. **Thin Client:** Light read only OS
6. **Thick Client:** Applications installed locally on thick client



Study notes

Acronym: Desktops **T-A-W**; PCs **D-E-D-S-Thin-Thick** from dumb to thick.

Trail: A building-up ladder: a dumb screen, then emulation, then a PC without a disk, a full PC, a thin client, a thick client.

If you forget: Each step gives the terminal more of its own.

Link: Q4 desktop and notebook definitions.

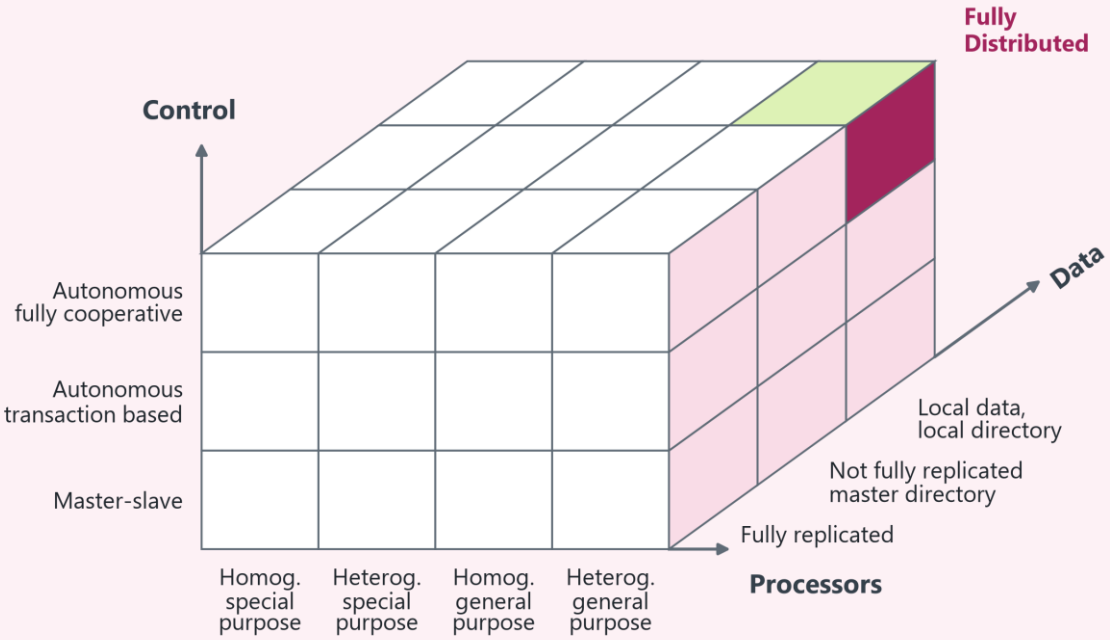
Test yourself: What does a thin client run?

6. Draw and explain Enslow model for distributed processing systems

A:

Distributed system = distributed hardware + distributed control + distributed data

- A system could be classified as a distributed system if all three categories (hardware, control, data) reach a certain degree of decentralization.



Control:

1. Single fixed control point.
2. **Single dynamic control point:** In multiple CPU cases the controller changes from time to time among CPUs.
3. **A fixed master/slave structure:** In a system with one CPU and one co-processor, the CPU is a fixed master and the co-processor is a fixed slave.
4. **A dynamic master/slave structure:** The role of master/slave is modifiable by software.
5. **Multiple homogeneous control points:** Copies of the same controller are used.
6. **Multiple heterogeneous control points:** Different controllers are used.

Data:

1. Centralized databases with a single copy of both files and directory.
2. Distributed files with a single centralized directory and no local directory.
3. Replicated database with a copy of files and a directory at each site.
4. Partitioned database with a master that keeps a complete duplicate copy of all files.
5. Partitioned database with a master that keeps only a complete directory.
6. Partitioned database with no master file or directory.

Study notes

Acronym: **H + C + D**; control 1 to 6, data 1 to 6.

Trail: A cube: hardware, control and data each go from one point to fully spread; the far corner is a full DCS.

If you forget: Every list goes from one central thing to many independent ones.

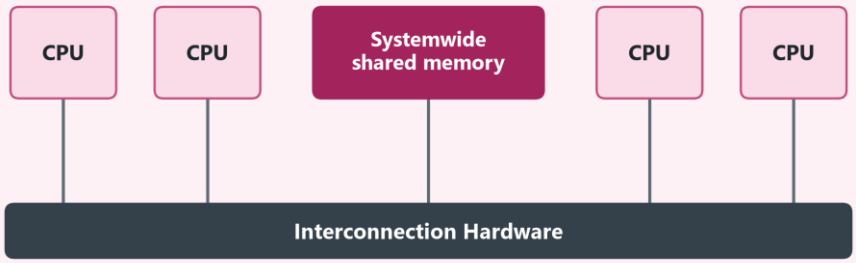
Test yourself: When is a system classified as distributed in the Enslow model?

7. Explain Tightly Coupled Multiprocessor System?

A:

Tightly coupled systems: They are multiprocessor systems. They are a population of processors sharing addressing to memory and devices.

- **Sharing:** It is accomplished by utilizing single OS.
- Tightly coupled systems are referred to as parallel processing systems or shared memory, where a single system wide primary memory (address space) is shared by all the processors.
- If any processor writes the value 100 to the memory location x, any other processor subsequently reading from location x will get the value 100.
- Any communication between the processors takes place by the shared memory.
- The number of processors that can be deployed is small and limited by the bandwidth of the shared memory.



Study notes

Acronym: **One OS, one memory**

Trail: Roommates with one fridge: everybody sees the same milk, but the kitchen door limits how many fit.

If you forget: Shared memory explains every line, including the limit.

Link: Q8 is the opposite; Q9 compares them.

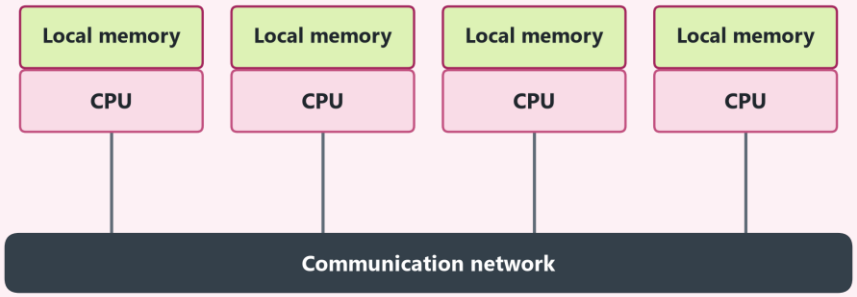
Test yourself: What limits the number of processors?

8. Explain Loosely Coupled Multiprocessor System?

A:

Loosely Coupled Systems: The loose interconnection of computer systems comes very close to the concept of distributed processing by the attribute of multiple complete systems.

- Systems connected over I/O pathways traditionally distinct from networks.
- Many of these pathways differ from tightly coupled systems only in the physical sense that they do not share memory or operating systems.
- They may be functionally quite dependent, and as a result take on the characteristics of a single node, a single 'system' rather than an interconnection of two nodes.



Study notes
Acronym: **I/O, no sharing, single node**
Trail: Neighbours who pass notes through the door: separate houses but they act like one family.
If you forget: Complete systems that do not share memory or OS.
Link: Q7 is the opposite; Q10 adds networks.
Test yourself: What do loosely coupled systems not share?

9. Differentiate between Loosely Coupled and Tightly Coupled Multiprocessor Systems

A:

Aspect	Loosely Coupled	Tightly Coupled
1 · Foundation		
Data rate	Loosely Coupled Multiprocessor System has low data rate.	Tightly coupled multiprocessor system has high data rate.
2 · Access and resources		
Memory	There is distributed memory in loosely coupled multiprocessor system.	There is shared memory, in tightly coupled multiprocessor system.
Connection to I/O devices	In loosely coupled multiprocessor, there is direct connection between processor and I/O devices.	While in tightly coupled multiprocessor, IOPIN helps connection between processor and I/O devices.
3 · Management and protection		
Memory conflicts	In loosely coupled multiprocessor, Memory conflicts don't take place.	While tightly coupled multiprocessor system has memory conflicts.
4 · Multi-user safety net		
Interaction between tasks	Loosely Coupled Multiprocessor system has low degree of interaction between tasks.	Tightly Coupled multiprocessor system has high degree of interaction between tasks.
Applications	Applications of loosely coupled multiprocessor are in DCSs.	Applications of tightly coupled multiprocessor are in parallel processing systems.

Study notes

Acronym: **D-M-C | M | I-A**

Trail: Speed first (data rate), then where memory lives and how I/O connects, then what goes wrong (conflicts), then the people side (task interaction, applications).

If you forget: Tightly coupled is always the fast, shared, high-interaction column.

Link: Q7 and Q8.

Test yourself: Which system has memory conflicts?

10. Explain networking distributed Multiprocessor System?

A:

Networked Systems: Loosely coupled systems using networks are referred to as DCS.

1 · Foundation

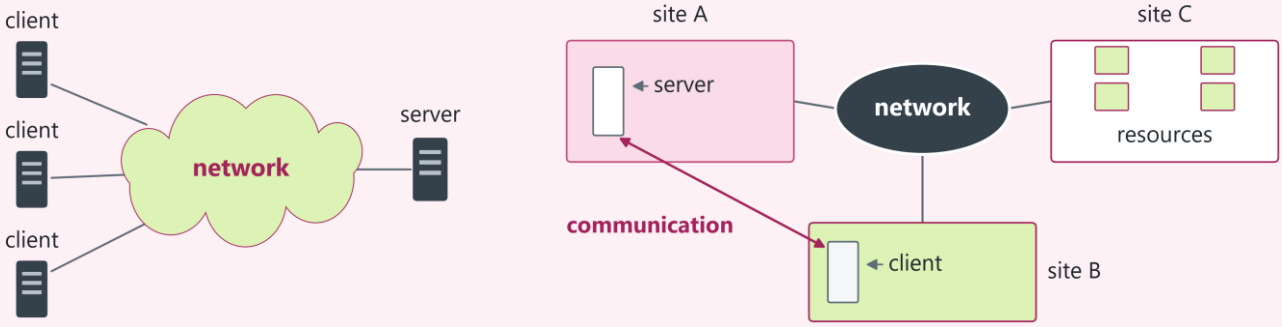
- Processors can be located far from each other to cover a wider geographical area, DCS are more freely expandable and can have an almost unlimited number of processors.

2 · Access and resources

- The processors do not share memory, and each processor has its own local memory.
- If a processor writes the value 100 to the memory location x, this write operation will only change the contents of its local memory and will not affect the contents of the memory.
- For a particular processor, its own resources are local, whereas the other processors and their resources are remote.
- A processor and its resources are referred to as a node or site or machine of the DCS.

3 · Communication

- All physical communication between the processors is done by passing messages across the interconnection network.



Study notes

Acronym: **Far | Local, write, local/remote, node | Messages**

Trail: Offices in many cities: they can be anywhere, each keeps its own files, and they talk only by mail.

If you forget: No shared memory, so only messages.

Link: Q8 loosely coupled systems; Q7's write-100 example is the opposite.

Source note: The slide line ends "the contents of the memory": it means the other processors' memories.

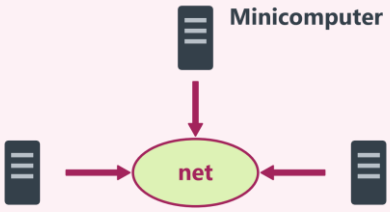
Test yourself: How is all physical communication done?

11. Explain minicomputer distributed Multiprocessor System?

A:

Minicomputer model: It is a simple extension of the centralized time-sharing system.

- A DCS consists of a few minicomputers interconnected by a communication network where each minicomputer usually has multiple users simultaneously logged on to it.
- Several interactive terminals are connected to each minicomputer.
- The network allows a user to access remote resources that are available on some machine other than the one on to which the user is currently logged.
- The minicomputer model may be used when resource sharing with remote users is desired.
- Early Internet is an example of a minicomputer DCS.



Drawbacks of Minicomputer Model:

1 · Foundation

- Very expensive.

2 · Users and tasks

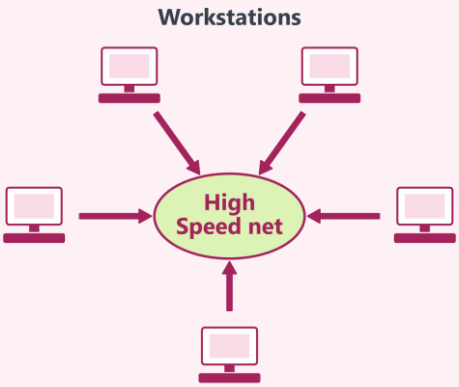
- Each user is logged on to one specific minicomputer, with remote access to other minicomputers.
- The processing of tasks is done by the processors of a single minicomputer.

> **Study notes**
> **Acronym:** Drawbacks **$-U-T**
> **Trail:** An old time-sharing club: each member is tied to one machine, the work stays on it, and it costs a lot.
> **If you forget:** Cost, user, task.
> **Link:** Q12 and Q13 are the next models.
> **Test yourself:** Give an example of a minicomputer DCS.

12. Explain workstation distributed Multiprocessor System?

A:

Workstation Model (WS Model): It consists of network of powerful PCs, each one with its own hard disk and local filesystem, interconnected over high speed network.



- A DCS based on the WS model consists of several WSs interconnected by a communication network.
- An organization may have several WSs located throughout an infrastructure where each WS is equipped with its own disk & serves as a single-user computer.
- At any one time a significant proportion of the WSs are idle which results in the waste of large amounts of CPU time.
- **Idea:** To interconnect all these WSs by a high-speed LAN so that idle WSs are used to process jobs of users who are logged onto other WSs & do not have sufficient processing power at their own WSs to get their jobs processed efficiently.
- **Example:** Sprite system & Xerox PARC

Study notes

Acronym: **WS = single user, idle = reuse**

Trail: Desks in an office at lunch: idle desks run other people's jobs over a fast LAN.

If you forget: The idea line is the answer.

Link: Q13 adds servers; Q14 pools the processors.

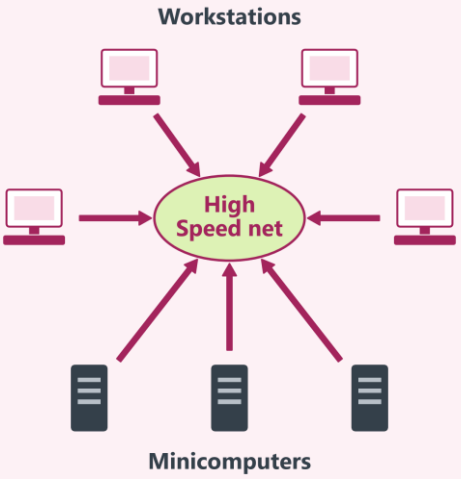
Test yourself: Which systems are the examples of the workstation model?

13. Explain workstation server distributed Multiprocessor System?

A:

Workstation-Server Model (WS-Server Model): It consists of multiple workstations coupled with powerful servers with extra hardware to store the file systems and other software like databases

- The WS model is a network of DISK-FULL PC WS having its own disk & a local file system.
- A DCS based on the WS-server model consists of a few minicomputers and several WSs interconnected by a network.
- A user logs onto a WS called his own PC.
- Normal computation activities required by the user's processes are performed at the user's PC.
- Requests for services provided by special servers are sent to a server providing that type of service that performs the user's requested activity & returns the result of request processing to the user's workstation.



- The user's processes do not need to be migrated to the server machines for getting the work done by those machines.

Study notes

Acronym: **WS + Server**

Trail: Cook at home, but order from a bakery for bread: the bakery returns the result, you don't move into it.

If you forget: Normal work local, special work on servers, processes stay home.

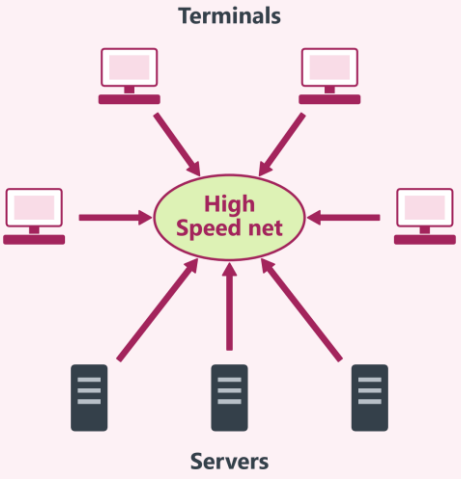
Link: Q12 and Q14.

Test yourself: Where is normal computation done in the WS-server model?

14. Explain Multiprocessor pool distributed Systems?

A:

Processor-pool Model: It consists of multiple processors: a pool of processors and a group of workstations



- The processor-pool model is based on the observation that most of the time a user does not need any computing power but once in a while the user may need a very large amount of computing power for a short time.
- Unlike the WS-SERVER model in which a processor is allocated to each user, in processor-pool model the processors are pooled together to be shared by the users as needed.
- The pool of processors consists of a large number of microcomputers & minicomputers attached to the network.
- Each processor in the pool has its own memory to load & run a system program or an application program of the distributed computing system.
- No home machine is present & the user does not log onto any machine.
- **Advantages:** Better utilization of processing power & greater flexibility.
- **Example:** Cambridge Distributed Computing System.

Study notes

Acronym: **Pool, no home**

Trail: A taxi rank: you take a car only when you need one, nobody owns a car.

If you forget: Short bursts of large computing power shared on demand.

Link: Q13 allocates a processor per user; here they are pooled.

Test yourself: What are the advantages of the processor-pool model?

GOOD LUCK

Quick review

Q	Question	Remember
1	Explain the technologies that move the computer systems from centralized to distributed.	M-L-P-M, many networked computers.
2	Mention the roles of microprocessor/minicomputer technology in the evolution of DCS?	1/1000th price, 64-bit, multicore, horizontal market, HCI.
3	Mention the roles of local area and wide area networks technology in the evolution of DCS?	LAN: building, fast. WAN: earth, 10^4-10^9 bps.
4	Mention the roles of the PC technology in the evolution of DCS?	Smartphone = full computer; plug, desktop, notebook.
5	Mention the different applications/configurations of the PC in the evolution of DCS?	Tower, all-in-one, workstation; dumb to thick.
6	Draw and explain Enslow model for distributed processing systems	Hardware + control + data decentralized.
7	Explain Tightly Coupled Multiprocessor System?	Shared memory, single OS, parallel processing.
8	Explain Loosely Coupled Multiprocessor System?	Multiple complete systems, no shared memory.
9	Differentiate between Loosely Coupled and Tightly Coupled Multiprocessor Systems	Loose: distributed memory, low rate. Tight: shared, high.
10	Explain networking distributed Multiprocessor System?	Loosely coupled + network; local memory, messages.
11	Explain minicomputer distributed Multiprocessor System?	Extension of time-sharing; expensive.
12	Explain workstation distributed Multiprocessor System?	Idle WSs process others' jobs.
13	Explain workstation server distributed Multiprocessor System?	Local work at the PC, services on servers.
14	Explain Multiprocessor pool distributed Systems?	Shared pool, no home machine; Cambridge.